

TEXT MINING

05th Dec 2022

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INTRODUCTION

- The university needed advice on how to improve its curriculum because they were looking to do it by looking at present day market situations.
- It has reached business analytics students to provide them with some insights.
- We accessed the Co-op services of the institutions as business analytics students and collected information about students and employers from 2014 to 2020.

DATA

**2014 TO
2020**

Student's and Employer's
data from Co-op services
of University

Goals, Co-op experience,
Improvements personally
and from university

**STUDENT'S
DATA**

**EMPLOYER'S
DATA**

Student's strengths,
weakness, Areas of
development, improvements
in University curriculum

Numbers of
responses from 7
years

10000+

DATA ANALYSIS - TEXT MINING

UNSTRUCTURED DATA

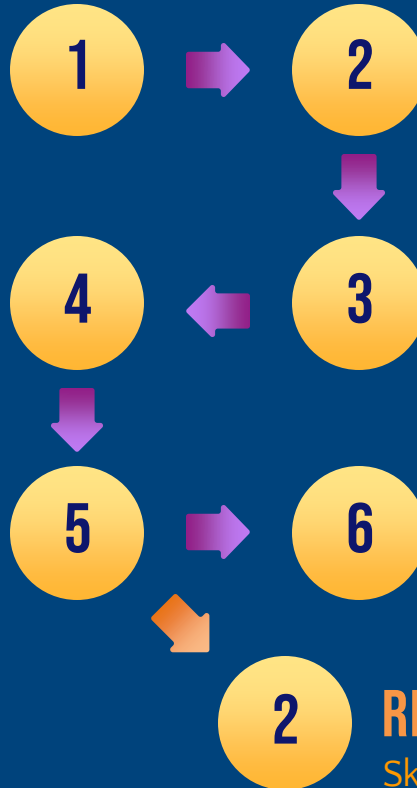
Co-op Data of students & Employee's from year 2014 to 2020

REMOVE EXTRA SPACES

Once we remove stop words, numbers and symbols there are lot blank spaces in the text

FREQUENCY OF WORDS & STEMMING

Finding number of times each word has repeated



REMOVE STOP WORDS

Removing stop words like 'I', 'me' which are not important in analyzing

REMOVE NUMBERS & SYMBOLS

Numbers and symbols don't interpret anything that helps students, university or employers

BI-GRAMS

Forming a meaningful 2 word sentence with words from frequency

REMOVE STOP WORDS, ALPHABETS

Skip to step 2 if we find any unwanted stop words for better data processing

ANALYSIS OF STUDY GOALS

- The most commonly observed skill gaps were in MS Excel, Communication skills, and Technical skills.
- Here, we discovered that the majority of students had optimistic attitudes about their goals and experienced sensations of eagerness, delight, and trust.



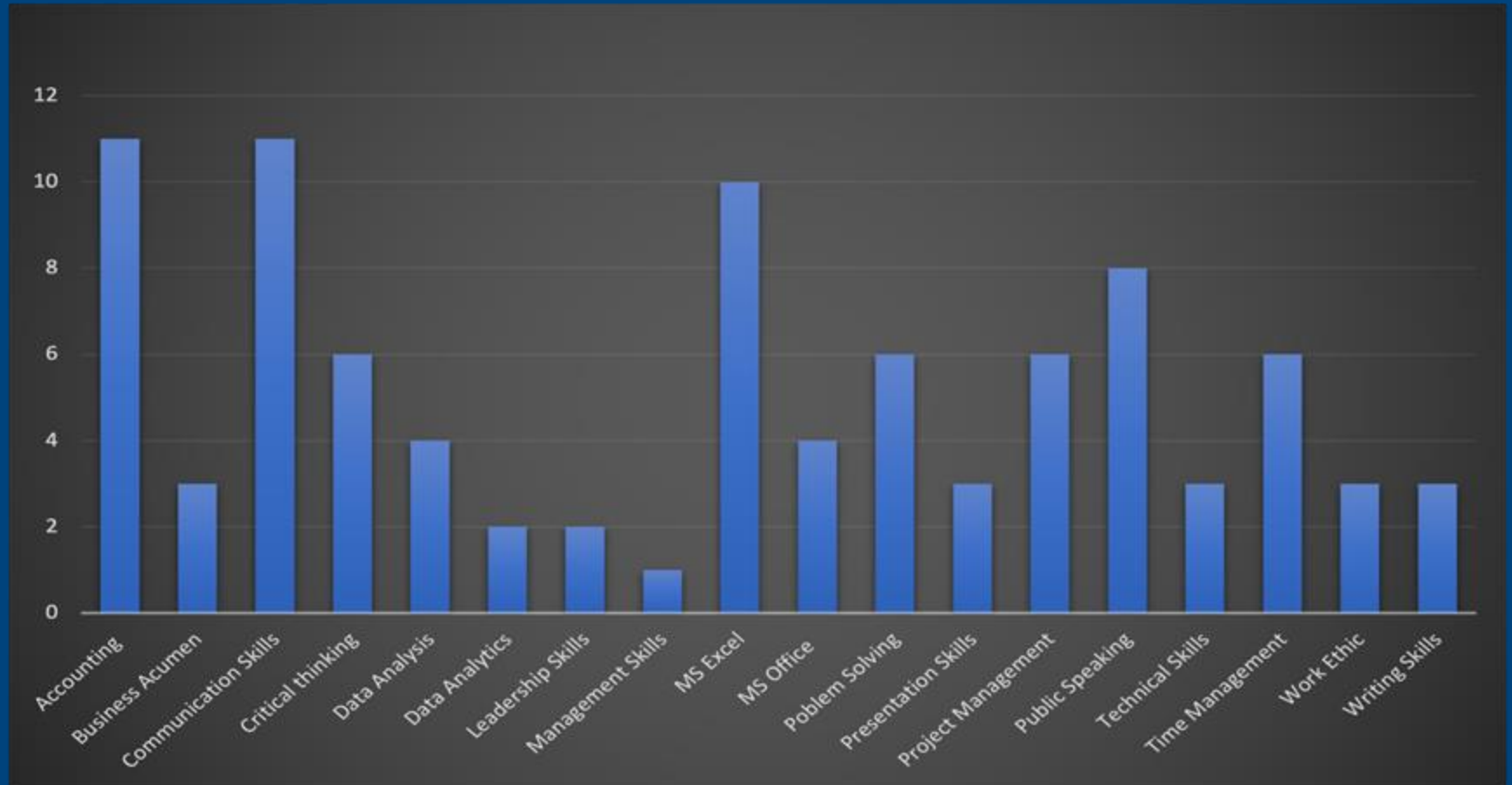
ANALYSIS OF STUDY CLASSROOM ACTIVITIES

- The most commonly observed classroom activities were in MS Excel, Communication skills, and Technical skills.
- Many students were focusing on things like project management, teamwork, excel abilities, and presentation skills.



ANALYSIS EMPLOYER'S RESPONSES

SUGGESTED SKILLS TO IMPROVE BY EMPLOYER



ANALYSIS OF EMPLOYER'S RESPONSES

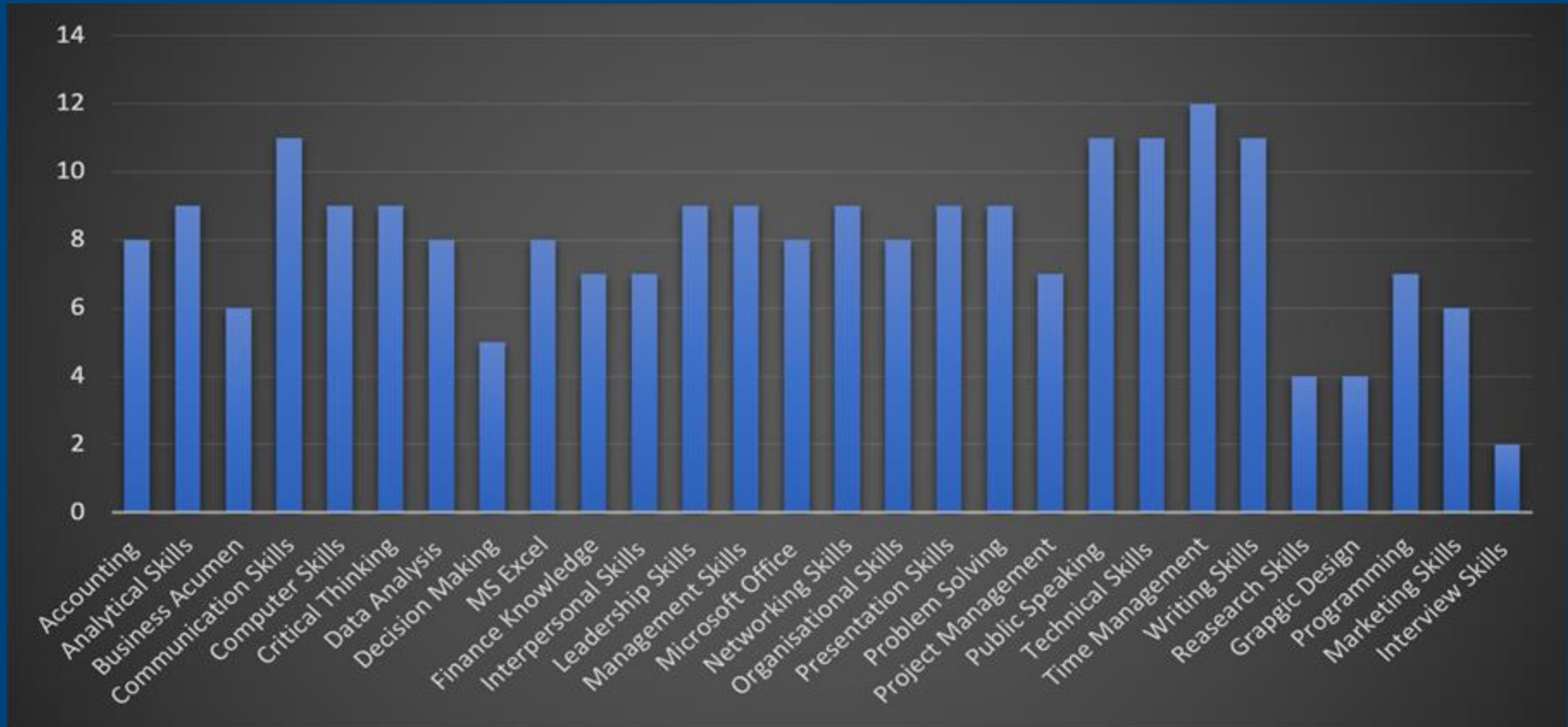
SUGGESTED SKILLS TO IMPROVE BY EMPLOYER

- Across the years it was observed that there was a larger skill gap as far as soft skills were concerned as compared to technical skills.
- The most commonly observed skill gaps were in MS Excel, Communication skills, and Accounting.



ANALYSIS OF STUDENT'S REQUIREMENT

FROM COLLEGE AFTER CO-OP



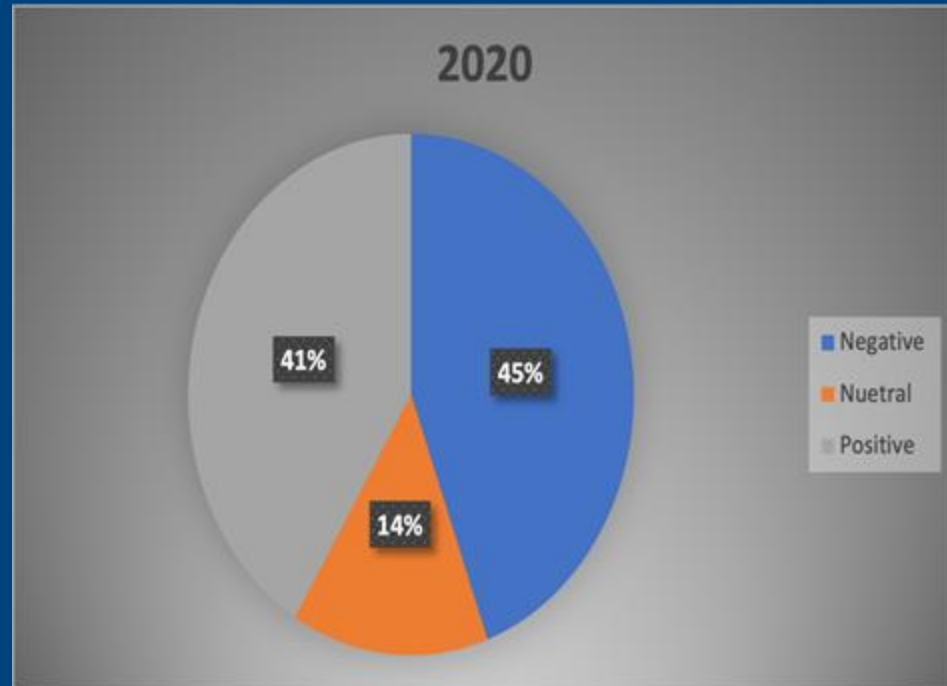
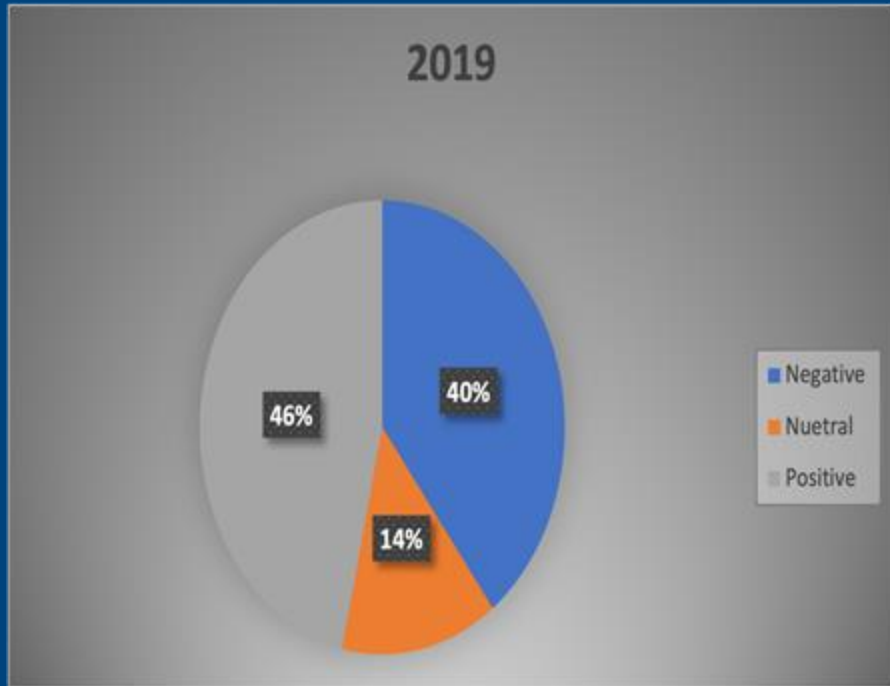
ANALYSIS OF STUDENT'S REQUIREMENT

FROM COLLEGE AFTER CO-OP

- After the Co-Op, most of the common requirements were in Microsoft Office, communication skills, and data analysis.
- We discovered that phrases like writing acumen, project, computer, and management were often utilized in the students' post-co-op requirements.



SENTIMENT ANALYSIS OF EMPLOYER'S RESPONSE - WEAKNESSES



LIMITATIONS

- We have only used comments from the past two years for classroom activities and student goals that students anticipate once their co-op is through, by considering how the market has changed during 2014 to 2018.
- Tri-grams can also be utilized for better analysis to gain more insights, however for this presentation, we have only used bi-grams and frequency (stemming).
- We used Azure Machine Learning in Excel for sentiment analysis, which just provided us with the positive or negative and percentage to indicate the strength of the sentiment. However, it didn't give us the phrases (such as avoid or fail for a negative) that led it to select the sentiment. For a more thorough examination, we can employ different sentiment analysis techniques.

CONCLUSION

- After analyzing the data from student goals, we discovered that investment banking, wealth management, financial accounting, digital marketing, and data analyst were the most typical goals specified by students.
- Many students found Excel projects, Financial accounting and Business analytics classes really helped them during their co-op.
- Employers across the years have suggested that students lacked in skills such as Advance Excel, Communication skills, Time management, Accounting and Public speaking.
- Employers reported a larger skill gap in soft skills than the technical skills.
- On analysing the data through the years we found that students want to improve their communication skills, technical skills, time management, writing skills and public speaking.
- It was observed that majority of the skills that employers found students were lacking in, were very similar to skill gaps that students faced in their co-op, and wanted to develop after co-op.
- On performing Sentiment analysis on employer's response for student weakness we found that the responses with positive sentiment were almost equal to those with negative sentiments.

Thank You!

